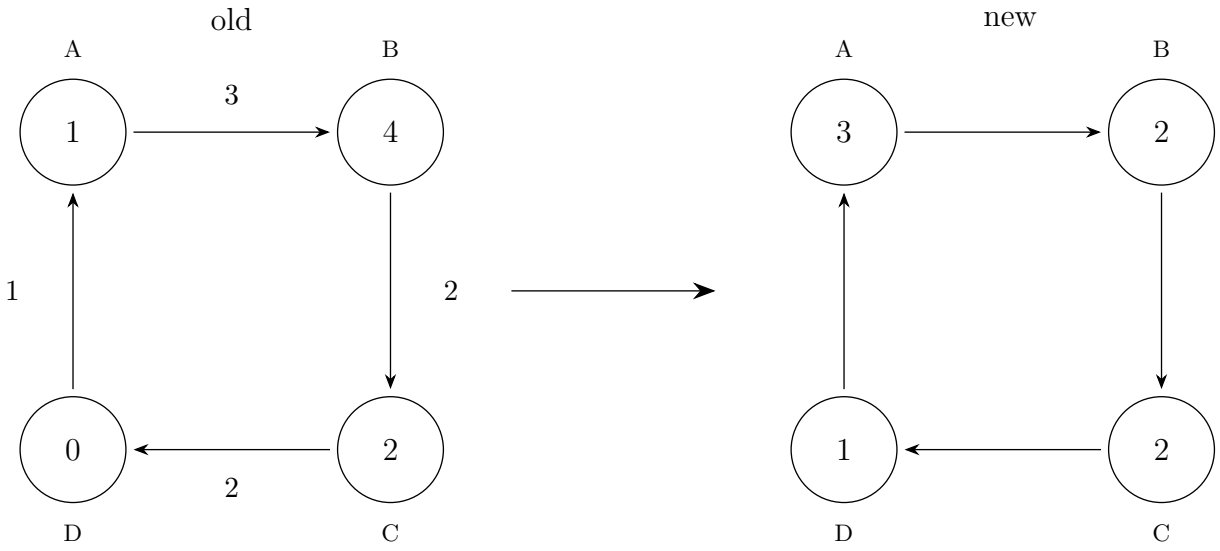
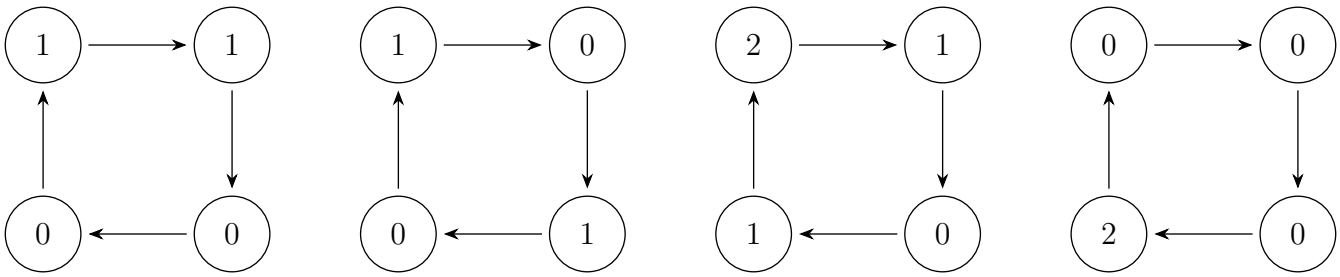
Use whole-number heights, including 0. Follow each arrow: the gap between its two old heights becomes the new height at its tail. Make every new tower before changing any old tower. An all-zero start takes 0 moves.

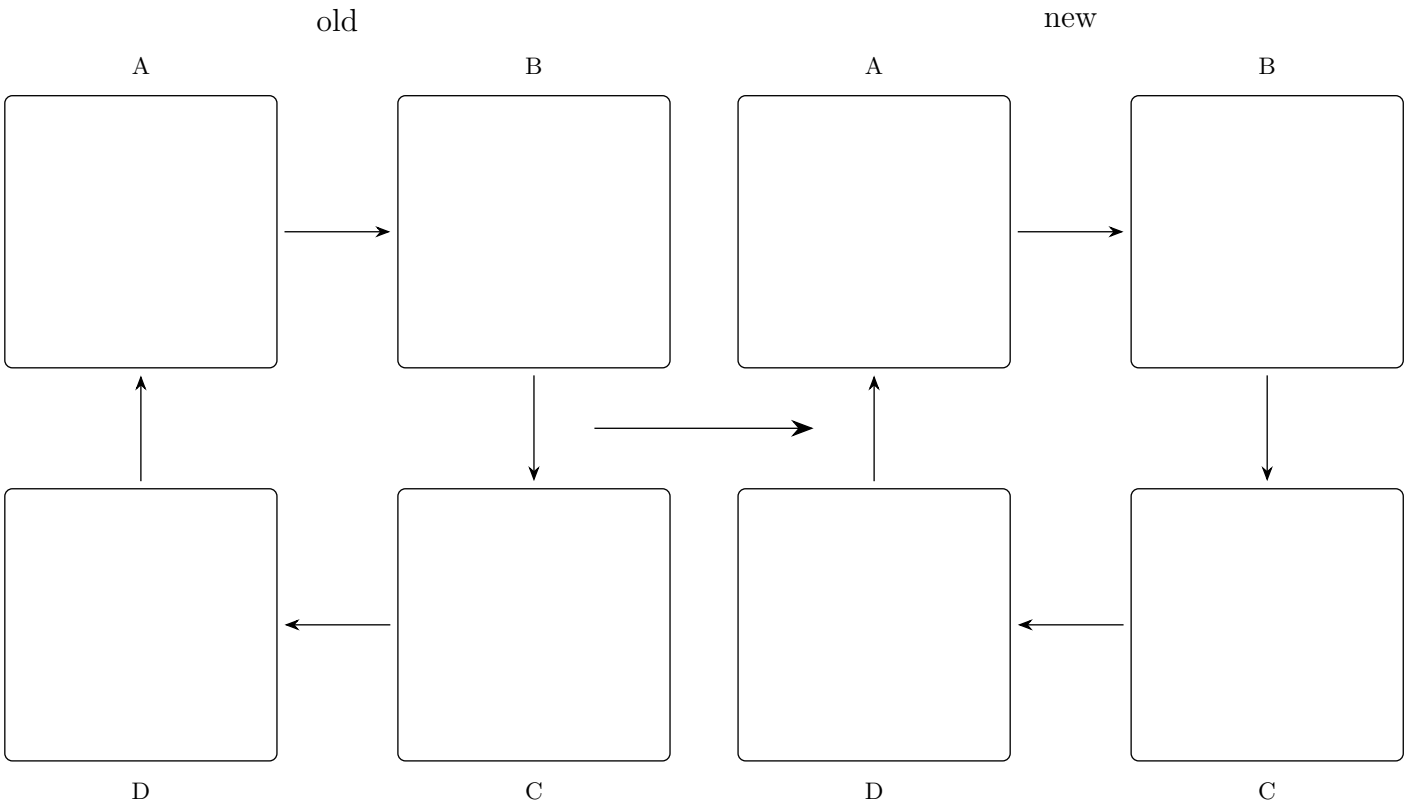


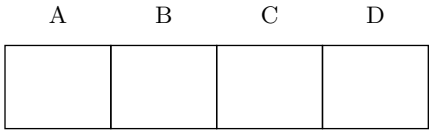
Edge A to B has gap 3, so new A is 3.

Problem 1: Build these starts and keep making new rings. Stop when all four towers are 0.

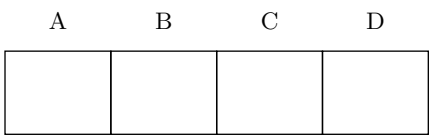


Problem 2: Choose heights from 0 to 4. Find a start that lasts as many moves as you can; count the move that makes all four towers 0.



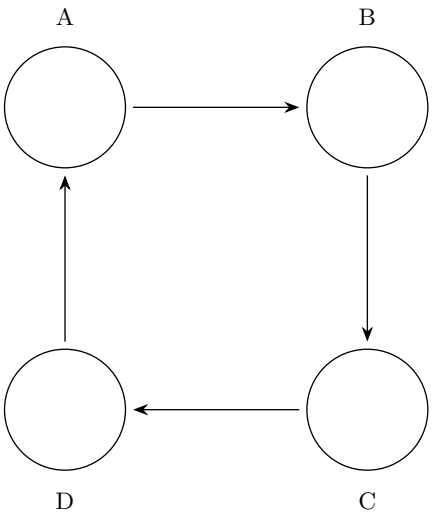


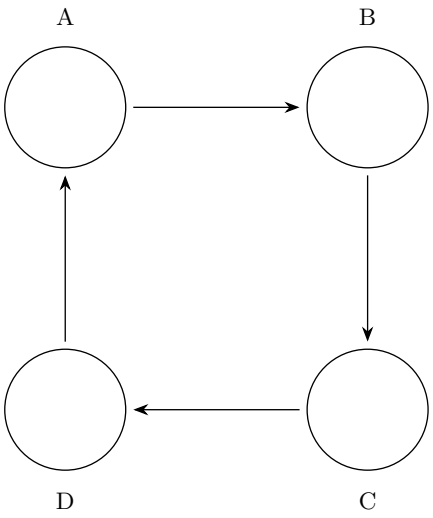
Moves to zero: _____



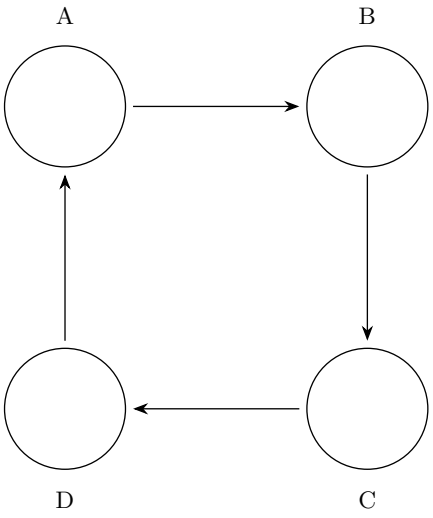
Moves to zero: _____

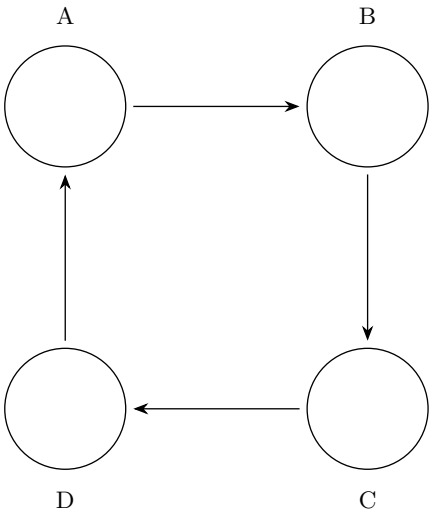
Problem 3: Find different starts that become all zeros in one move.



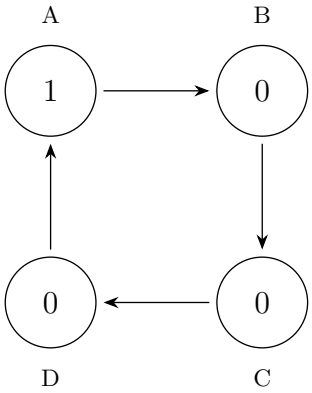


Problem 4: Find different starts that first become all zeros on the second move.

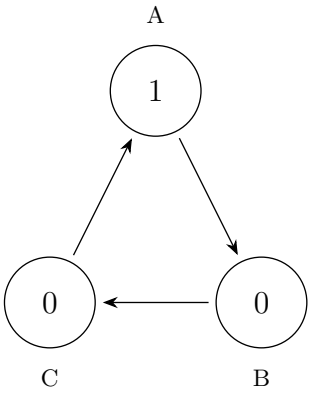




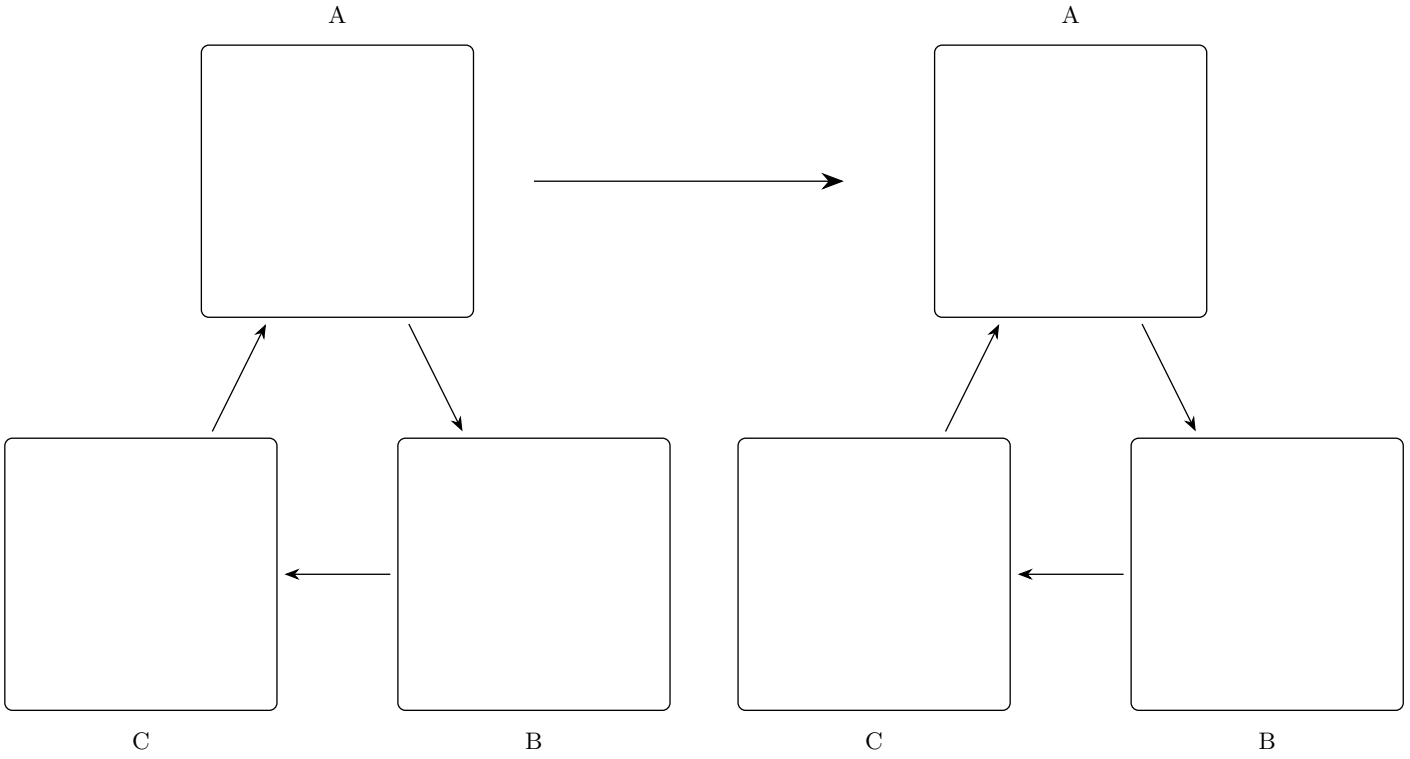
Problem 5: Try both starts with the same gap rule. Stop at all zeros or when the whole ring repeats.

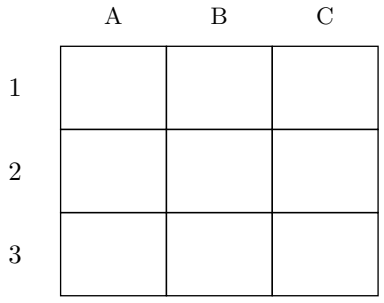


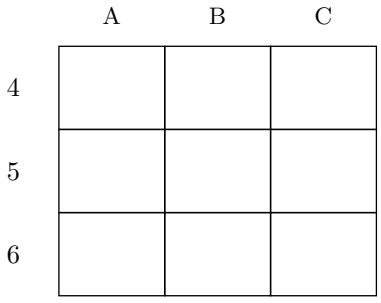
old



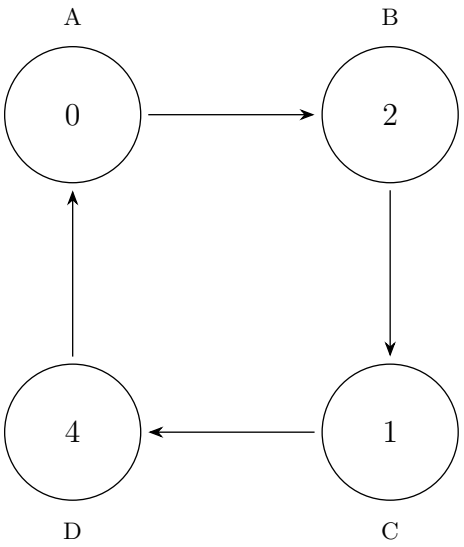
new

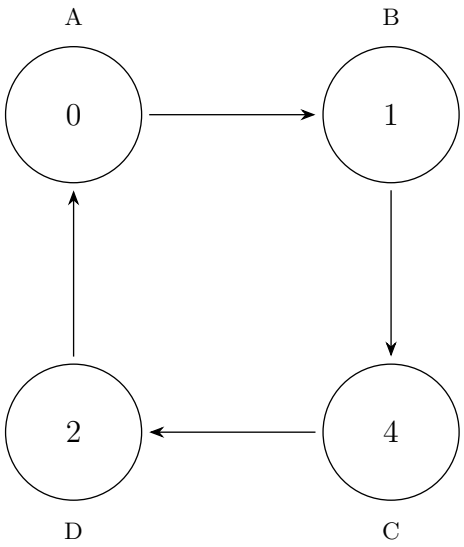


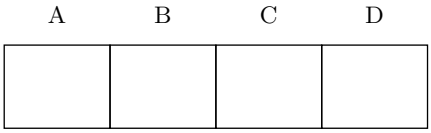




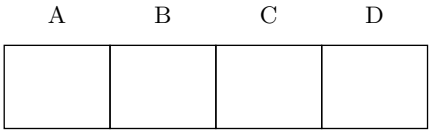
Problem 6: Put 0, 1, 2, and 4 around the four stations. Find an order that lasts longer than the other orders you try.



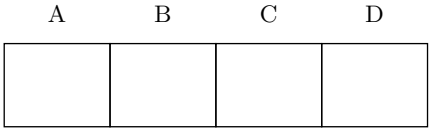




Moves to zero: _____



Moves to zero: _____



Moves to zero: _____